23 Northern Bank has two types of account.
Both accounts pay compound interest.

Cash savings account	**Shares account**
Interest	Interest
2.5% per annum	3.5% per annum

Ali invests £2000 in the cash savings account.
Ben invests £1600 in the shares account.

(a) Work out who will get the most interest by the end of 3 years.
You must show all your working.

Ali, year 1: £2000 × 1.025 = £2050

1 mark
Calculate one year of interest

Ali, after 3 years: £2000 × 1.025^3

= £2153.78 (to the nearest penny)

1 mark
Compound for three years

Ali's interest = £2153.78 – £2000 = £153.78

Ben: £1600 × 1.035^3 = £1773.95

1 mark
Complete both compound interest calculations

Ben's interest = £1773.95 – £1600 = £173.95

Ben gets more interest: £173.95 > £153.78.

1 mark
Compare interest gained, not the final balances

(4)

In the 3rd year the rate of interest for the shares account is changed to 4% per annum.

(b) Does this affect who will get the most interest by the end of 3 years?
Give a reason for your answer.

No. Ben's interest becomes £182.52:

£1600 × 1.035^2 × 1.04 – £1600 = £182.5184.

This is still greater than Ali's £153.78.

1 mark
Conclusion remains Ben, supported by the new comparison

(1)

(Total for Question 23 is 5 marks)

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